STAFF REVIEW · H446 218 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. The pivot is already in its final sorted position.

2. A connection between vertices with an associated cost or distance.

3. 0.

4. 8.

I Do — source worked steps

1. Predict the next state — P gives tentative Q=3 and R=7.

2. Reveal and explain — Settle Q=3, improving R to 5 and discovering S=9.

3. Reveal and explain — Settle R=5, improving S to 6.

4. Reveal and explain — Predecessors S←R←Q←P give P→Q→R→S, cost 6.

We Do — guided CHECK

1. Reveal and explain
From O, N improves from 5 to 3 and P gets a tentative 10.
From N, P improves to 6; P is next. Route M→O→N→P costs 6.

You Do — source mark schemes

1. [3] A=0 [1], B=4 [1], C=2 [1]. C is chosen next.

2. [6] After S: A3, B7 [1]; settle A3 [1] improving B to5 and T to11 [2]; settle B5, T improves to7 [1]; route S→A→B→T, cost 7 [1].

3. [5] Dijkstra assumes non-negative edges [1], so no later path can improve a settled vertex [1]; a negative edge can invalidate that assumption [1]. Tentative distances can change [1], but settled distances are final under the precondition [1].

Exit Ticket — source answer

1. False. It chooses the unsettled vertex with the smallest total tentative distance from the start.